

Deep_Learning

October 26, 2024

0.1 Easy record-linkage using the LinkTransformer library

From https://colab.research.google.com/drive/1OqUB8sqpUvrnC8oa_1RoOUzV6DaAKL4N?usp=sharing#scroll

0.2 Setup

```
[1]: # An environment with python=3.11 works. (3.12 does not work.)
```

```
#!pip install linktransformer
```

```
[2]: import linktransformer as lt
import pandas as pd
import os
```

```
[3]: os.chdir('C:/Dropbox/Applied-Economics/task12')
```

1 Supported Pre-trained models

We support several LinkedTransformer models, check out our guide on selecting the best models on this tutorial notebook [model zoo](#). Our framework supports all language models on [HuggingFace](#). We highly recommend the sentence transformers models also available on HuggingFace. For more on sentence-transformers, visit their [website](#)

For users of LinkedTransformer who are new to NLP, here is a guide we have prepared for you to select an appropriate semantic similarity model. (<https://colab.research.google.com/drive/1SAvQdgYiX2CinoTC8Y5qtKScNwx3DfYXf?usp=sharing>)

For the use case of linking companies and harmonising trade products (applications covered in our paper), here is a tutorial on how to do it with a high level interface, check out our package [repo](#)

1.1 Standard (deep) merge on a key

Just like you would in any merge command!

```
[4]: ###Read data
df1 = pd.read_csv("toy_comp_1.csv")
df2 = pd.read_csv("toy_comp_2.csv")

df1.head(3)
```

```
[4]: CompanyID      CompanyName      Industry      Founded_Year      Country
0         1          TechCorp      Technology          2005          USA
1         2      InfoTech Solutions      Technology          1998          Canada
2         3      GlobalSoft Inc          Software          2010          India
```

```
[5]: import linktransformer as lt

# Test your function here
df_lm_matched = lt.merge(df2, df1, merge_type='1:m', on="CompanyName",
    ↪model="all-MiniLM-L6-v2", left_on=None, right_on=None)

df_lm_matched.head(10)
```

2024-10-26 08:24:23 - Load pretrained SentenceTransformer: all-MiniLM-L6-v2

c:\Users\chadi\anaconda3\envs\islp\Lib\site-packages\huggingface_hub\file_download.py:797: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(

2024-10-26 08:24:25 - Use pytorch device_name: cpu

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[5]: CompanyID_x      CompanyName_x      Revenue (Millions USD) \
0         1          Tech Corporation          5000
1         2      InfoTech Soln          4500
2         3      GlobalSoft Incorporated          3000
3         4      DataTech Corporation          2500
4         5          SoftSys Limited          4000
5         6          TechCorp          5500
6         8      AlphaSoft Systems          3800

      Num_Employees      Country_x      id_lt_x      CompanyID_y      CompanyName_y \
0         10000          USA          0          1          TechCorp
1         8500          Canada          1          2      InfoTech Solutions
2         6000          India          2          3      GlobalSoft Inc
3         5000          Germany          3          4          DataTech Co
4         7500          UK          4          5          SoftSys Ltd
5         12000          USA          5          1          TechCorp
6         7000          Spain          6          7          AlphaSoft

      Industry      Founded_Year      Country_y      id_lt_y      score
0      Technology          2005          USA          0      0.828613
1      Technology          1998          Canada          1      0.806597
2      Software          2010          India          2      0.941341
```

3	Data Analytics	2012	Germany	3	0.934297
4	Software	2003	UK	4	0.811891
5	Technology	2005	USA	0	1.000000
6	Software	2007	France	6	0.783905

```
[6]: df_lm_matched[["CompanyName_x", "CompanyName_y", "score", "Revenue (Millions_
↳USD)", "Num_Employees"]]
```

```
[6]:      CompanyName_x      CompanyName_y      score \
0      Tech Corporation      TechCorp      0.828613
1      InfoTech Soln      InfoTech Solutions      0.806597
2      GlobalSoft Incorporated      GlobalSoft Inc      0.941341
3      DataTech Corporation      DataTech Co      0.934297
4      SoftSys Limited      SoftSys Ltd      0.811891
5      TechCorp      TechCorp      1.000000
6      AlphaSoft Systems      AlphaSoft      0.783905
```

	Revenue (Millions USD)	Num_Employees
0	5000	10000
1	4500	8500
2	3000	6000
3	2500	5000
4	4000	7500
5	5500	12000
6	3800	7000

1.2 Merge with Blocking - easily in a line

Often it is necessary to merge within partitions of datasets using a “blocking” variable. In this case, we merge “CompanyName” with only companies that have the same “Country” value.

```
[7]: df1 = pd.read_csv("toy_comp_1.csv")
df2 = pd.read_csv("toy_comp_2.csv")

# Test your function here
df_lm_matched = lt.merge_blocking(df2, df1, merge_type='1:m', on="CompanyName",
↳model="all-MiniLM-L6-v2", left_on=None, right_on=None,
↳blocking_vars=["Country"])
df_lm_matched.head(10)
```

Number of blocks in df1: 6

Number of blocks in df2: 6

2024-10-26 08:24:25 - Load pretrained SentenceTransformer: all-MiniLM-L6-v2

c:\Users\chadi\anaconda3\envs\islp\Lib\site-

packages\huggingface_hub\file_download.py:797: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(

2024-10-26 08:24:27 - Use pytorch device_name: cpu

Merging block UK

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Merging block USA

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Merging block India

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Merging block Germany

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Merging block Canada

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[7]: CompanyID_x      CompanyName_x  Revenue (Millions USD) \
0      5.0      SoftSys Limited      4000.0
1      1.0      Tech Corporation      5000.0
2      6.0      TechCorp      5500.0
3      3.0      GlobalSoft Incorporated      3000.0
4      4.0      DataTech Corporation      2500.0
5      2.0      InfoTech Soln      4500.0
6      NaN      NaN      3800.0
7      NaN      NaN      NaN
```

```
      Num_Employees Country_x  id_lt_x  CompanyID_y      CompanyName_y \
0      7500.0      UK      0.0      5.0      SoftSys Ltd
1      10000.0      USA      0.0      1.0      TechCorp
2      12000.0      USA      1.0      1.0      TechCorp
3      6000.0      India      0.0      3.0      GlobalSoft Inc
4      5000.0      Germany      0.0      4.0      DataTech Co
5      8500.0      Canada      0.0      2.0      InfoTech Solutions
6      7000.0      NaN      NaN      NaN      NaN
7      NaN      NaN      NaN      NaN      NaN
```

```
      Industry  Founded_Year  Country_y  id_lt_y      score  CompanyID \
0      Software      2003.0      UK      0.0  0.811891      NaN
1      Technology      2005.0      USA      0.0  0.828613      NaN
```

2	Technology	2005.0	USA	0.0	1.000000	NaN
3	Software	2010.0	India	0.0	0.941341	NaN
4	Data Analytics	2012.0	Germany	0.0	0.934297	NaN
5	Technology	1998.0	Canada	0.0	0.806597	NaN
6	NaN	NaN	NaN	NaN	NaN	8.0
7	Software	2007.0	NaN	NaN	NaN	7.0

	CompanyName	Country
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN
5	NaN	NaN
6	AlphaSoft Systems	Spain
7	AlphaSoft	France

1.3 All functions also support multiple keys!

Matching over multiple keys is also supported, for example, CompanyName and ProductDescription. Multiple columns are serialized by concatenating with a token that the package selects to be suitable with the base language model tokenizer.

```
[8]: df1 = pd.read_csv("toy_multi_1.csv")
df2 = pd.read_csv("toy_multi_2.csv")

df1.head(3)
```

```
[8]:   ProductID      CompanyName ProductCategory \
0         101          TechCorp      Electronics
1         102  InfoTech Solutions      Electronics
2         103   GlobalSoft Inc        Software

      ProductDescription  Price
0  High-performance laptop with SSD    1500
1   Smartphone with AI capabilities    1000
2  Enterprise CRM software for businesses    2000
```

```
[9]: # Test your function here
df_lm_matched = lt.merge(df2, df1, merge_type='1:m', on=["CompanyName",
↳ "ProductDescription"], model="all-MiniLM-L6-v2", left_on=None, right_on=None)

df_lm_matched.head(10)
```

2024-10-26 08:24:27 - Load pretrained SentenceTransformer: all-MiniLM-L6-v2

c:\Users\chadi\anaconda3\envs\islp\Lib\site-packages\huggingface_hub\file_download.py:797: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when

possible. If you want to force a new download, use `force\_download=True`.
warnings.warn(

2024-10-26 08:24:28 - Use pytorch device_name: cpu

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[9]: ProductID_x      CompanyName_x ProductCategory_x \
0      201      TechCorp      Electronics
1      202  InfoTech Solutions      Electronics
2      203      GlobalSoft      Software
3      204      DataTech      Data Analysis
4      205      SoftSys      Software
5      206      TechCorp      Electronics
6      207  InfoTech Solutions      Electronics
7      208      GlobalSoft      Software
8      209      DataTech      Data Analysis
9      210      SoftSys      Software
```

```
ProductDescription_x Price_x id_lt_x \
0 High performance laptop with solid state drive      1400      0
1      Smartphone with AI support      950      1
2      CRM software for enterprises      1800      2
3      Data analytics platform for large datasets      1700      3
4      Document management software for businesses      2000      4
5      Powerful gaming laptop with SSD      1600      5
6      Smartphone with advanced AI capabilities      1100      6
7      Advanced CRM software for businesses      2100      7
8      Big data analytics platform for enterprises      1900      8
9      Enterprise document management system      2300      9
```

```
ProductID_y      CompanyName_y ProductCategory_y \
0      101      TechCorp      Electronics
1      102  InfoTech Solutions      Electronics
2      103      GlobalSoft Inc      Software
3      104      DataTech Co      Data Analytics
4      105      SoftSys Ltd      Software
5      101      TechCorp      Electronics
6      102  InfoTech Solutions      Electronics
7      103      GlobalSoft Inc      Software
8      104      DataTech Co      Data Analytics
9      105      SoftSys Ltd      Software
```

```
ProductDescription_y Price_y id_lt_y      score
0      High-performance laptop with SSD      1500      0 0.847999
1      Smartphone with AI capabilities      1000      1 0.963210
2      Enterprise CRM software for businesses      2000      2 0.926065
```

3	Data analysis platform for big data	1800	3	0.812184
4	Document management system for enterprises	2200	4	0.918345
5	High-performance laptop with SSD	1500	0	0.904424
6	Smartphone with AI capabilities	1000	1	0.967993
7	Enterprise CRM software for businesses	2000	2	0.937032
8	Data analysis platform for big data	1800	3	0.882190
9	Document management system for enterprises	2200	4	0.915138

1.4 Quick Aggregation or Classification as KNN retrieval

A merge can be used to aggregate or classify (KNN) - say fine product descriptions to coarser categories.

```
[10]: df_coarse = pd.read_csv("coarse.csv")
      df_fine = pd.read_csv("fine.csv")

[11]: df_lm_aggregate = lt.aggregate_rows(df_fine, df_coarse,
      ↪model="all-mpnet-base-v2", left_on="Fine Category Name", right_on="Coarse_
      ↪Category Name")

      df_lm_aggregate.head()
```

2024-10-26 08:24:29 - Load pretrained SentenceTransformer: all-mpnet-base-v2

c:\Users\chadi\anaconda3\envs\islp\Lib\site-packages\huggingface_hub\file_download.py:797: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(

2024-10-26 08:24:31 - Use pytorch device_name: cpu

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[11]: Fine Category ID Fine Category Name Coarse Category ID_x id_lt_x \
0          101          Laptops          1          0
1          102 Desktop Computers          1          1
2          103      Smartphones          1          2
3          104          Tablets          1          3
4          105      Headphones          1          4

      Coarse Category ID_y Coarse Category Name id_lt_y score
0          1 Computers & Electronics          0 0.596126
1          1 Computers & Electronics          0 0.551886
2          1 Computers & Electronics          0 0.540363
3          1 Computers & Electronics          0 0.352265
4          1 Computers & Electronics          0 0.351324
```

```
[12]: df_lm_aggregate[["Fine Category Name", "Coarse Category Name"]].head(10)
```

```
[12]: Fine Category Name      Coarse Category Name
0          Laptops  Computers & Electronics
1  Desktop Computers  Computers & Electronics
2      Smartphones  Computers & Electronics
3          Tablets  Computers & Electronics
4      Headphones  Computers & Electronics
5      Speakers  Computers & Electronics
6  Refrigerators      Home Appliances
7  Washing Machines      Home Appliances
8   Vacuum Cleaners      Home Appliances
9          Toasters      Home Appliances
```

1.5 Bypass translation with Cross-lingual merges

You can merge across languages - without having the need to translate anything! Just use a multilingual model to merge.

```
[13]: df_french = pd.read_csv("translation_1.csv")
df_english = pd.read_csv("translation_2.csv")

df_french.head(3)
```

```
[13]: ProductID      Libellé du Produit      Catégorie      Prix
0      101      Ordinateur portable performant      Électronique      1400
1      102  Smartphone avec prise en charge de l'IA      Électronique      950
2      103      Logiciel CRM pour entreprises      Logiciel      1800
```

```
[14]: # French to English Merge

# Test your function here
df_lm_matched = lt.merge(df_french, df_english, merge_type='1:m',
    ↪left_on="Libellé du Produit", right_on="Product Label",
    ↪model="distiluse-base-multilingual-cased-v1")

df_lm_matched.head(10)
```

2024-10-26 08:24:31 - Load pretrained SentenceTransformer: distiluse-base-multilingual-cased-v1

```
c:\Users\chadi\anaconda3\envs\islp\Lib\site-
packages\huggingface_hub\file_download.py:797: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
  warnings.warn(
```

2024-10-26 08:24:33 - Use pytorch device_name: cpu

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[14]: ProductID_x                               Libellé du Produit \
0      101                                Ordinateur portable performant
1      102                        Smartphone avec prise en charge de l'IA
2      103                        Logiciel CRM pour entreprises
3      104  Plateforme d'analyse de données pour gros ense...
4      105                        Gestion de documents pour les entreprises
5      106                        Caméra de sécurité à haute résolution
6      107                                Imprimante tout-en-un
7      108                                Casque audio sans fil
8      109                        Système de divertissement à domicile
9      110                                Robot aspirateur intelligent
```

	Catégorie	Prix	id_lt_x	ProductID_y \
0	Électronique	1400	0	101
1	Électronique	950	1	102
2	Logiciel	1800	2	103
3	Analyse de données	1700	3	104
4	Logiciel	2000	4	105
5	Électronique	1200	5	106
6	Électronique	600	6	107
7	Électronique	300	7	108
8	Électronique	2500	8	109
9	Électronique	800	9	110

	Product Label	Category	Price	id_lt_y \
0	High-performance laptop	Electronics	1400	0
1	Smartphone with AI support	Electronics	950	1
2	CRM software for businesses	Software	1800	2
3	Data analytics platform for large datasets	Data Analytics	1700	3
4	Document management for businesses	Software	2000	4
5	High-resolution security camera	Electronics	1200	5
6	All-in-one printer	Electronics	600	6
7	Wireless headphones	Electronics	300	7
8	Home entertainment system	Electronics	2500	8
9	Intelligent robot vacuum cleaner	Electronics	800	9

	score
0	0.484677
1	0.746061
2	0.897755
3	0.934434
4	0.965277
5	0.955552
6	0.877312

```
7 0.722391
8 0.934203
9 0.821990
```

```
[15]: df_lm_matched[["Libellé du Produit", "Product Label", "Catégorie", "Category"]].
      ↪head(10)
```

```
[15]:
```

	Libellé du Produit \		Product Label		Catégorie \
0	Ordinateur portable performant		High-performance laptop		Électronique
1	Smartphone avec prise en charge de l'IA		Smartphone with AI support		Électronique
2	Logiciel CRM pour entreprises		CRM software for businesses		Logiciel
3	Plateforme d'analyse de données pour gros ense...		Data analytics platform for large datasets		Analyse de données
4	Gestion de documents pour les entreprises		Document management for businesses		Logiciel
5	Caméra de sécurité à haute résolution		High-resolution security camera		Électronique
6	Imprimante tout-en-un		All-in-one printer		Électronique
7	Casque audio sans fil		Wireless headphones		Électronique
8	Système de divertissement à domicile		Home entertainment system		Électronique
9	Robot aspirateur intelligent		Intelligent robot vacuum cleaner		Électronique

	Category
0	Electronics
1	Electronics
2	Software
3	Data Analytics
4	Software
5	Electronics
6	Electronics
7	Electronics
8	Electronics
9	Electronics

1.6 Perform clustering at the row level in one line!

Several algorithms are supported and more to be added soon.

```
[16]: df=pd.read_csv("toy_comp_2.csv")
df_cluster=lt.cluster_rows(df,on="CompanyName",model="sentence-transformers/
↳all-MiniLM-L6-v2",cluster_type= "agglomerative",
cluster_params= {'threshold': 0.7})
df_cluster.head(10)
```

2024-10-26 08:24:33 - Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2

c:\Users\chadi\anaconda3\envs\islp\Lib\site-packages\huggingface_hub\file_download.py:797: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(

2024-10-26 08:24:35 - Use pytorch device_name: cpu

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[16]: CompanyID      CompanyName  Revenue (Millions USD)  Num_Employees \
0          1      Tech Corporation           5000           10000
1          2      InfoTech Soln             4500           8500
2          3  GlobalSoft Incorporated         3000           6000
3          4      DataTech Corporation         2500           5000
4          5      SoftSys Limited             4000           7500
5          6          TechCorp                5500          12000
6          8      AlphaSoft Systems           3800           7000
```

```
Country  cluster
0      USA      0
1  Canada      5
2   India      3
3  Germany      4
4       UK      2
5      USA      0
6   Spain      1
```

1.7 Deduplicate your rows in one line!

Deduplicate using a language model with one line of code! Several clustering algorithms like SLINK, HDBSCAN and DBSCAN are supported

```
[17]: df=pd.read_csv("toy_comp_2.csv")
df_dedup=lt.dedup_rows(df,on="CompanyName",model="sentence-transformers/
↳all-MiniLM-L6-v2",cluster_type= "agglomerative",
cluster_params= {'threshold': 0.7})
df_dedup.head(10)
```

2024-10-26 08:24:35 - Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2

```
c:\Users\chadi\anaconda3\envs\islpl\Lib\site-
packages\huggingface_hub\file_download.py:797: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
warnings.warn(
```

```
2024-10-26 08:24:36 - Use pytorch device_name: cpu
```

```
Deduplicating dataframe with originally 7 rows
```

```
Checking for and dropping exact duplicates
```

```
Number of rows after dropping exact duplicates: 7
```

```
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
```

```
Number of rows after deduplication: 6
```

```
[17]:
```

	CompanyID	CompanyName	Revenue (Millions USD)	Num_Employees \
0	1	Tech Corporation	5000	10000
1	2	InfoTech Soln	4500	8500
2	3	GlobalSoft Incorporated	3000	6000
3	4	DataTech Corporation	2500	5000
4	5	SoftSys Limited	4000	7500
6	8	AlphaSoft Systems	3800	7000

```
Country
0    USA
1  Canada
2   India
3  Germany
4     UK
6   Spain
```

```
[18]: df.head(10)
```

```
[18]:
```

	CompanyID	CompanyName	Revenue (Millions USD)	Num_Employees \
0	1	Tech Corporation	5000	10000
1	2	InfoTech Soln	4500	8500
2	3	GlobalSoft Incorporated	3000	6000
3	4	DataTech Corporation	2500	5000
4	5	SoftSys Limited	4000	7500
5	6	TechCorp	5500	12000
6	8	AlphaSoft Systems	3800	7000

```
Country
0    USA
1  Canada
2   India
3  Germany
4     UK
5     USA
```

1.8 We also support getting semantic similarity scores between values of specified columns!

All you need is a dataframe containing the columns of interest as keys, and you are done! This will be a very powerful drop in replacement for edit distance calculations

```
[19]: df=pd.read_csv("toy_pairs.csv")
      df.head(3)
```

```
[19]:      company_name_1      company_name_2  label
0  Acme Corporation      ACME Corporation      1
1   TechGiant Inc.      TechGiant Ltd.      0
2   Widgets R Us  Widgets Incorporated      0
```

```
[20]: df_eval=lt.evaluate_pairs(df, model="sentence-transformers/
      ↪all-MiniLM-L6-v2",left_on="company_name_1",right_on="company_name_2",openai_key=None)
      df_eval.head()
```

2024-10-26 08:24:36 - Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2

c:\Users\chadi\anaconda3\envs\islp\Lib\site-packages\huggingface_hub\file_download.py:797: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(

2024-10-26 08:24:37 - Use pytorch device_name: cpu

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[20]:      company_name_1      company_name_2  label      score
0  Acme Corporation      ACME Corporation      1  1.000000
1   TechGiant Inc.      TechGiant Ltd.      0  0.906331
2   Widgets R Us  Widgets Incorporated      0  0.780843
3  SuperTech Solutions      SuperTech Corp      1  0.758345
4   Globex Industries      Globex Corp      1  0.918245
```

1.9 Get all pair-wise distances between keys from two datasets

```
[21]: df=pd.read_csv("toy_pairs.csv")
      df_eval=lt.all_pair_combos_evaluate(df, model="sentence-transformers/
      ↪all-MiniLM-L6-v2",left_on="company_name_1",right_on="company_name_2",openai_key=None)
      df_eval.head(100)
```

2024-10-26 08:24:37 - Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2

```
c:\Users\chadi\anaconda3\envs\islpl\Lib\site-
packages\huggingface_hub\file_download.py:797: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
warnings.warn(
```

```
2024-10-26 08:24:39 - Use pytorch device_name: cpu
```

```
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
```

```
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
```

```
[21]:
```

	left_on	right_on	score
0	Acme Corporation	ACME Corporation	1.000000
1	Acme Corporation	TechGiant Ltd.	0.451013
2	Acme Corporation	Widgets Incorporated	0.184706
3	Acme Corporation	SuperTech Corp	0.514488
4	Acme Corporation	Globex Corp	0.458063
..	...	...	...
95	BestWare Inc.	Innovations Unlimited	0.353746
96	BestWare Inc.	AlphaTech Systems	0.292267
97	BestWare Inc.	ABC Solutions	0.060783
98	BestWare Inc.	TechMaster Corp.	0.517866
99	BestWare Inc.	BestWare Ltd.	0.885602

```
[100 rows x 3 columns]
```

1.10 Get K nearest matches for each row in the left df

```
[22]: df1 = pd.read_csv("toy_comp_1.csv")
df2 = pd.read_csv("toy_comp_2.csv")

df_lm_matched_2nn = lt.merge_knn(df2, df1, on="CompanyName",
    model="sentence-transformers/all-MiniLM-L6-v2", left_on=None, right_on=None,
    k=2)

df_lm_matched_2nn.head()
```

```
2024-10-26 08:24:39 - Load pretrained SentenceTransformer: sentence-
transformers/all-MiniLM-L6-v2
```

```
c:\Users\chadi\anaconda3\envs\islpl\Lib\site-
packages\huggingface_hub\file_download.py:797: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
warnings.warn(
```

```
2024-10-26 08:24:40 - Use pytorch device_name: cpu
```

```
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
```

```
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
```

Adding embeddings to index

Searching index

LM matched on key columns - left: CompanyName_x, right: CompanyName_y

```
[22]:
```

	CompanyID_x	CompanyName_x	Revenue (Millions USD)	\
0	1	Tech Corporation	5000	
1	1	Tech Corporation	5000	
2	2	InfoTech Soln	4500	
3	2	InfoTech Soln	4500	
4	3	GlobalSoft Incorporated	3000	

	Num_Employees	Country_x	id_lt_x	CompanyID_y	CompanyName_y	\
0	10000	USA	0	6	TechCorp	
1	10000	USA	0	1	TechCorp	
2	8500	Canada	1	2	InfoTech Solutions	
3	8500	Canada	1	4	DataTech Co	
4	6000	India	2	3	GlobalSoft Inc	

	Industry	Founded_Year	Country_y	id_lt_y	score
0	Technology	2005	USA	5	0.828613
1	Technology	2005	USA	0	0.828613
2	Technology	1998	Canada	1	0.806597
3	Data Analytics	2012	Germany	3	0.641552
4	Software	2010	India	2	0.941341

```
[23]: df_lm_matched = lt.merge(df1, df2, merge_type='1:m', on="CompanyName",  
    ↳model="dell-research-harvard/lt-wikidata-comp-en", left_on=None,  
    ↳right_on=None)  
  
df_lm_matched.head(10)
```

Warning: Keys in df1 are not unique

2024-10-26 08:25:45 - Load pretrained SentenceTransformer: dell-research-harvard/lt-wikidata-comp-en

c:\Users\chadi\anaconda3\envs\islpl\Lib\site-packages\huggingface_hub\file_download.py:797: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.
warnings.warn(
 ↳

2024-10-26 08:25:47 - Use pytorch device_name: cpu

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Batches: 0%| | 0/1 [00:00<?, ?it/s]

```
[23]:
```

	CompanyID_x	CompanyName_x	Industry	Founded_Year	Country_x	\
0	1	TechCorp	Technology	2005	USA	
1	2	InfoTech Solutions	Technology	1998	Canada	

2	3	GlobalSoft Inc	Software	2010	India
3	4	DataTech Co	Data Analytics	2012	Germany
4	5	SoftSys Ltd	Software	2003	UK
5	6	TechCorp	Technology	2005	USA
6	7	AlphaSoft	Software	2007	France

	id_lt_x	CompanyID_y	CompanyName_y	Revenue (Millions USD)	\
0	0	6	TechCorp	5500	
1	1	2	InfoTech Soln	4500	
2	2	3	GlobalSoft Incorporated	3000	
3	3	4	DataTech Corporation	2500	
4	4	5	SoftSys Limited	4000	
5	5	6	TechCorp	5500	
6	6	8	AlphaSoft Systems	3800	

	Num_Employees	Country_y	id_lt_y	score
0	12000	USA	5	1.000000
1	8500	Canada	1	0.707271
2	6000	India	2	0.991263
3	5000	Germany	3	0.991782
4	7500	UK	4	0.997345
5	12000	USA	5	1.000000
6	7000	Spain	6	0.971633

```
[24]: #save
df_lm_matched.to_csv("toy_comp_matched.csv",index=False)
```